

National University of Computer & Emerging Sciences

Homework # 2(a)

Counting

1. From a group of n people, suppose that we want to choose a committee of k , $k \leq n$, one of whom is to be designated as chairperson.
 - a. By focusing first on the choice of the committee and then on the choice of the chair, argue that there are $\binom{n}{k} k$ possible choices.
 - b. By focusing first on the choice of the non-chair committee members and then on the choice of the chair, argue that there are $\binom{n}{k-1} (n - k + 1)$ possible choices.
 - c. By focusing first on the choice of the chair and then on the choice of the other committee members, argue that there are $n \binom{n-1}{k-1}$ possible choices.
 - d. Conclude from parts (a), (b) and (c) that
$$\binom{n}{k} k = \binom{n}{k-1} (n - k + 1) = n \binom{n-1}{k-1}$$
by using the factorial definition.
2. If a multiple-choice test consists of 5 questions, each with 4 possible answers of which only 1 is correct,
 - a. in how many ways can a student check off one answer to each question?
 - b. in how many ways can a student check off one answer to each question and get all the answers wrong?
3.
 - a. How many distinct permutations can be made from the letters of the word COLUMNS?
 - b. How many of these permutations start with the letter M?
4.
 - a. Determine the coefficient of $x^9 y^3$ in the expansion of $(x + 2y)^{12}$.
 - b. Determine the coefficient of xyz^2 in the expansion of $(x + y + 3z)^4$.
5. Consider the following program segment, where i , j and k are integer variables.

```
For  $i := 1$  to 20 do
  For  $j := 1$  to  $i$  do
    For  $k := 1$  to  $j$  do
      print ( $i * j + k$ )
```

How many times is the print statement executed in this program segment?